Akihiro Miki

Project Assistant Professor

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Summary

- a Project Assistant Professor at the University of Tokyo, working on the intersection of biological understanding and intelligent machine design, with a focus on biomimetic robotics to investigate biological systems and their underlying principles.

Education

PhD **The University of Tokyo**, Mechano-Informatics — Tokyo, Japan, Apr 2023 – Mar 2026
- PhD in Graduate School of Information Science and Technology
- Dissertation: A Cross-Hierarchical Constructive Framework for Biological Understanding Based on Reproducing Emergent Functional Processes via Biomimetic Tissue Robotics

Master **The University of Tokyo**, Interdisciplinary Information Studies — Tokyo, Japan, Apr 2021 – Mar 2023
- Completed Master's program at Graduate School of Interdisciplinary Information Studies

Bachelor **The University of Tokyo**, Mechano-Informatics — Tokyo, Japan, Apr 2017 – Mar 2021
- Graduated from Department of Mechano-Informatics, Faculty of Engineering

M.D. Program (withdrawn) **Ehime University**, Medicine — Ehime, Japan, Mar 2017

Experience

The University of Tokyo, Project Assistant Professor — Tokyo, Japan, Apr 2026 – present, 1 month
Working at JSK Robotics Laboratory (Creative Informatics, Information Science and Technology).

JSPS (Japan Society for the Promotion of Science), (in japanese 日本学術振興会), JSPS Research Fellowships for young students (DC2) — Tokyo, Japan, Apr 2024 – Mar 2026, 2 years 1 month

Japan Science and Technology Agency (JST), The University of Tokyo "Advanced Human Resource Development Leading Green Transformation (GX) (SPRING GX)" Project student — Tokyo, Japan, Apr 2023 – Mar 2024, 1 year 1 month
- Description 1. JST Support for Pioneering Research Initiated by the Next Generation (SPRING) Program

The University of Tokyo, Research Assistant — Tokyo, Japan, May 2022 – Mar 2026, 4 years

Awards

Valedictorian (Graduation Ceremony Representative) Mar 2026
Selected to deliver the commencement address on behalf of graduating students.
Akihiro Miki
www.u-tokyo.ac.jp/focus/en/articles/z1301_00062.html

in Japanese, 2024 年度計測自動制御学会 学術奨励賞 Mar 2025
人の関節潤滑機能を模した人体模倣ロボットの液体滲出軟骨機構の構成法
三木章寛
www.sice.jp/info/info_press/press_20250512-2.html

in Japanese, 第 24 回計測自動制御学会システムインテグレーション部門講演会 優秀講演賞 Jan 2024
人の関節潤滑機能を模した人体模倣ロボットの液体滲出軟骨機構の構成法
三木章寛
sice-si.org/si2023

in Japanese, 第 3 回ロボティクスシンポジア優秀研究・技術賞 Sept 2023
高応答性を備え多様なロボットの扱いを可能とするハードウェア抽象化デバイス制御プラットフォーム開発
三木章寛, 永松祐弥，河原塚健人，平岡直樹，岡田慧，稲葉雅幸
www.rsj.or.jp/info/awards/category/ykg_rs

Publications

Exploring the proprioceptive potential of joint receptors using a biomimetic robotic joint 2026
Category: Biological Understanding via Robotics | Abstract: This study builds a biomimetic robotic joint to evaluate how joint receptor arrangements can encode proprioceptive signals and contribute to constructive understanding of biological sensing mechanisms.
Akihiro Miki, Shun Hasegawa, Sota Yuzaki, Yuta Sahara, Yoshimoto Ribayashi, Kento Kawaharazuka, Kei Okada
doi.org/10.1038/s41598-025-27311-3 (Scientific Reports | Vol. 16, No. 1, pp. 4724)

Development of a Hardware-abstracted Device Control Platform that Enables Handling of a Variety of Robots while Providing High Responsivity 2025
Category: Robot Systems | Abstract: This paper presents a high-responsivity hardware-abstracted device-control platform that standardizes heterogeneous robot hardware interfaces and enables robust operation across diverse robotic systems.
Akihiro Miki, Yuya Nagamatsu, Masahiro Bando, Kento Kawaharazuka, Yasunori Toshimitsu, Naoki Hiraoka, Kei Okada, Masayuki Inaba
doi.org/10.7210/jrsj.43.75 (Journal of the Robotics Society of Japan | Vol. 43, No. 1, pp. 75-86)

Designing Fluid-Exuding Cartilage for Biomimetic Robots Mimicking Human Joint Lubrication Function 2024
Category: Biomimetics | Abstract: We propose and validate a fluid-exuding artificial cartilage structure that reproduces key human joint lubrication behavior, improving durability and biomimetics functionality in robotic joint systems.
Akihiro Miki, Yuta Sahara, Kazuhiro Miyama, Shunnosuke Yoshimura, Yoshimoto Ribayashi, Kento Kawaharazuka, Shun Hasegawa, Kei Okada, Masayuki Inaba
doi.org/10.1109/robosoft60065.2024.10521920 (Proceedings of the 2024 IEEE-RAS International Conference on Soft Robotics | pp. 452-459)

System Architecture and Real-World Task Realization of Musculoskeletal Wheeled Robot Musashi-W with Various Hardware Components 2024

Category: Musculoskeletal Systems | Abstract: This work details the system architecture of the Musashi-W musculoskeletal wheeled robot and demonstrates integration of multiple hardware components for stable real-world task execution.

Akihiro Miki, Kento Kawaharazuka, Masahiro Bando, Kei Okada, Koji Kawasaki, Masayuki Inaba

doi.org/10.1007/978-3-031-44851-5_9 (Intelligent Autonomous Systems 18 | pp. 109-122)

Skills

Robotics

Languages

Japanese
Native speaker

English
Fluent

Interests

Robotics